

COS30018 – OPTION C

TASK 4 REPORT

Student Name: Syed Muhammad Mahdi

Student ID: 105172195

Date: 14 September 2025

1. Environment Setup:

Task 4 used the Python virtual environment of Task 2 and Task 3. It has TensorFlow, scikit-learn, pandas, numpy, matplotlib, and yfinance. Unimportant changes were done to make it compatible with Keras 3. The requirements file was stable and managed to address all the major dependencies.

2. Implementation of Task 4:

Task 4 was devoted to the experiment with three recurrent networks (LSTM, GRU, RNN) to predict stock prices. Models with various hyperparameters were trained in the script `task4_run.py`, according to a specified range of search space. Adaptive logging and rates of learning as well as early termination were introduced.

3. Testing and Results:

Apple (AAPL) stock data between 2018 and 2025 were used to execute the training process. The training lasted 32 epochs and a batch size of 32 in each set-up.

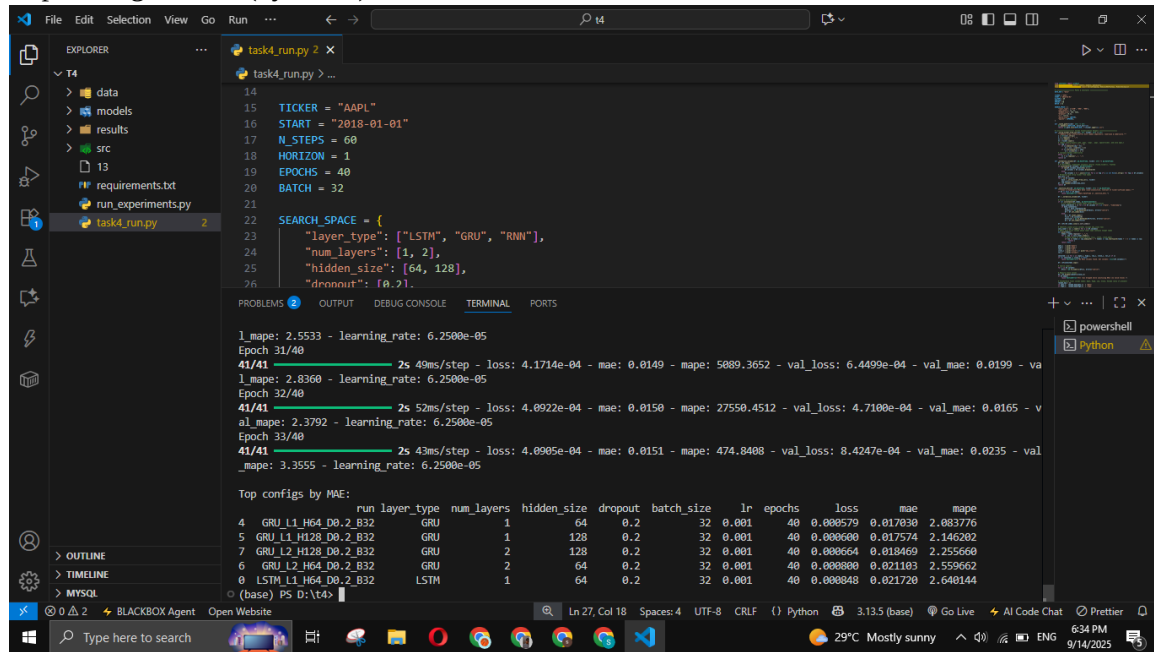
Sample logs from the console:

Epoch 31/40 – loss: 0.000417 – mae: 0.0149 – val_mae: 0.0199

Epoch 32/40 – loss: 0.000409 – mae: 0.0150 – val_mae: 0.0165

Epoch 33/40 – loss: 0.000409 – mae: 0.0151 – val_mae: 0.0235

Top Configurations (by MAE)



Run Layer Layers Hidden Dropout Batch LR Epochs Loss MAE MAPE.

4. Observations

GRU models took top spots on the leaderboard with the lowest MAE.

GRU with 1 layer, hidden size 64, dropout 0.2 (MAE 0.0173 on the best) was the best result.

- The metrics of validation were stable, which means that overfitting was not carried out.
- LSTM performed better in terms of MAE compared to GRU and was not a competitor to RNN configurations.

5. Conclusion

Task 4 built a hyperparameter search multistage stock prediction pipeline on several recurrent architectures. GRU was more competitive than LSTM and RNN in this case scenario, as it produced the best Mean Absolute Error. Experiment shows that GRUs are effective in time-series prediction of stock prices in terms of combined computation cost and accuracy.

Appendix

Screenshots included:

- The output of the training console with MAE and learning rate.
- Table of the highest configurations in order of MAE.
- Predicted and actual values (optional).